

目标函数：

$$E(x_q) = \frac{1}{2} \sum_{x \in D} K(d(x_q, x))(f(x) - \hat{f}(x))^2$$

线性模型：

$$\hat{f}(x) = \sum_{i=0}^n w_i a_i(x)$$

计算梯度：

$$\begin{aligned} \frac{\partial E}{\partial w_i} &= \frac{\partial}{\partial w_i} \left[\frac{1}{2} \sum_{x \in D} K(d(x_q, x))(f(x) - \hat{f}(x))^2 \right] \\ &= \sum_{x \in D} K(d(x_q, x))(f(x) - \hat{f}(x)) \frac{\partial}{\partial w_i} \left(f(x) - \sum_{j=0}^n w_j a_j(x) \right) \\ &= \sum_{x \in D} K(d(x_q, x))(f(x) - \hat{f}(x))(-a_i(x)) \\ &= - \sum_{x \in D} K(d(x_q, x))(f(x) - \hat{f}(x))a_i(x) \end{aligned}$$

梯度下降更新法则，

权重的变化量 Δw_i 定义为：

$$\Delta w_i = -\eta \frac{\partial E}{\partial w_i}$$

代入梯度：

$$\begin{aligned} \Delta w_i &= -\eta \left[- \sum_{x \in D} K(d(x_q, x))(f(x) - \hat{f}(x))a_i(x) \right] \\ &= \eta \sum_{x \in D} K(d(x_q, x))(f(x) - \hat{f}(x))a_i(x) \end{aligned}$$

将求和范围缩小至 k 个近邻，即得到原公式：

$$\Delta w_i = \eta \sum_{x \in k \text{ 的近邻}} K(d(x_q, x))(f(x) - \hat{f}(x))a_i(x)$$